

Disproof of conjecture 00000001097

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Verdict: FALSE

We disprove conjecture 00000001097 as stated. The refutation is unconditional and elementary. The mean root count of $x^d + ax + b$ over $\mathbb{F}_q$ is *exactly* 1 for every prime power q and every d ; the claimed “main term” is strictly larger than 1 and, when $\gcd(d-1, q-1) = 1$, coincides with the second moment $2 - 1/q$ rather than with the mean; and the conjectured variance $2 - 1/q$ is off by exactly 1. At $q = 5$, $d = 2$ the conjecture asserts both $9/5$ and 1 for the same mean, so it is internally inconsistent.

1 The conjecture

Definition 1 (as stated). The *mean root count* of the trinomial $x^d + ax + b$ over $\mathbb{F}_q$ is the expectation of the number of roots, taken over uniform $(a, b) \in \mathbb{F}_q^2$.

Conjecture. The mean equals $1 + (q-1)/q^{\gcd(d-1, q-1)}$ as the main term; when $\gcd(d-1, q-1) = 1$ the mean is exactly 1, and the variance of the mean is $2 - 1/q$.

Throughout, q is a prime power and $\mathbb{F}_q$ the field with q elements. For $(a, b) \in \mathbb{F}_q^2$ put

$$N(a, b) = \#\{x \in \mathbb{F}_q : x^d + ax + b = 0\},$$

so that the mean root count is

$$M(q, d) = \frac{1}{q^2} \sum_{a, b \in \mathbb{F}_q} N(a, b).$$

We write $g = \gcd(d-1, q-1)$.

2 The exact mean is 1

Theorem 2. For every prime power q and every integer $d \geq 1$,

$$M(q, d) = 1.$$

Proof. We count the triples $(a, b, x) \in \mathbb{F}_q^3$ satisfying $x^d + ax + b = 0$. By definition of M ,

$$q^2 M(q, d) = \#\{(a, b, x) \in \mathbb{F}_q^3 : x^d + ax + b = 0\}.$$

Fix $x \in \mathbb{F}_q$ and $a \in \mathbb{F}_q$. The map

$$\mathbb{F}_q \rightarrow \mathbb{F}_q, \quad b \mapsto x^d + ax + b$$

is a bijection: it is translation by the constant $x^d + ax$, with inverse $c \mapsto c - (x^d + ax)$. Hence it has exactly one zero, namely $b = -x^d - ax$. Consequently, for each of the q choices of x and each of the q choices of a there is exactly one b for which the equation holds. The number of triples is therefore $q \cdot q = q^2$, and

$$M(q, d) = \frac{q^2}{q^2} = 1. \quad \square$$

Only the field property of $\mathbb{F}_q$ is used, via invertibility of translation by b ; no hypothesis on d is needed.

Corollary 3. *For every prime power $q > 1$ and every d , the conjectured value*

$$1 + \frac{q-1}{q^{\gcd(d-1, q-1)}}$$

is strictly greater than 1, and therefore does not equal $M(q, d)$.

Proof. Since $q > 1$ we have $q-1 > 0$, and $q^g \geq q > 0$ for $g = \gcd(d-1, q-1) \geq 1$. Hence $(q-1)/q^g > 0$, so the displayed quantity is > 1 . By Theorem 2, $M(q, d) = 1$, so equality fails. $\square$

3 Internal contradiction at $q = 5$, $d = 2$

Theorem 4. *The conjecture is internally inconsistent.*

Proof. Take $q = 5$ and $d = 2$. Then

$$\gcd(d-1, q-1) = \gcd(1, 4) = 1.$$

The conjecture asserts, within one sentence, both

$$M(5, 2) = 1 + \frac{5-1}{5^1} = 1 + \frac{4}{5} = \frac{9}{5} \quad \text{and} \quad M(5, 2) = 1 \quad (\text{“exactly 1” when } \gcd(d-1, q-1) = 1).$$

Since $9/5 \neq 1$, these two clauses cannot both hold. The same argument applies to every $q > 1$ with $\gcd(d-1, q-1) = 1$: for instance $q = 7$, $d = 2$ gives $\gcd(1, 6) = 1$ and $1 + 6/7 = 13/7 \neq 1$. $\square$

Remark 5. The contradiction is between two clauses of the conjecture itself and does not require the exact computation of $M(5, 2)$: whatever the mean is, the conjecture demands two different values for it. Theorem 2 identifies the correct value, namely 1.

4 The variance clause is false

Let $N = N(a, b)$ for uniform $(a, b) \in \mathbb{F}_q^2$. We compute the exact second moment.

Lemma 6. *For every prime power q and every d ,*

$$\mathbb{E}[N^2] = 2 - \frac{1}{q}.$$

Proof. For each unordered pair $\{x, y\}$ of *distinct* points of $\mathbb{F}_q$, we count the pairs (a, b) for which both x and y are roots:

$$x^d + ax + b = 0, \quad y^d + ay + b = 0.$$

Subtracting the second equation from the first,

$$(x^d - y^d) + a(x - y) = 0.$$

Since $x \neq y$, the difference $x - y$ is nonzero and hence invertible in the field $\mathbb{F}_q$, so

$$a = -\frac{x^d - y^d}{x - y}$$

is uniquely determined; substituting back gives $b = -x^d - ax$, also unique. Thus each unordered pair of distinct points is a pair of common roots for exactly one (a, b) , and there are $\binom{q}{2} = q(q-1)/2$ such pairs.

Now sum $N(a, b)^2$ over all (a, b) : this counts ordered pairs (x, y) of roots of the same trinomial. Splitting into the diagonal $x = y$ and the off-diagonal $x \neq y$,

$$\sum_{a,b} N(a, b)^2 = \sum_{a,b} N(a, b) + 2 \cdot \#\{(a, b, \{x, y\}) : x \neq y, x, y \text{ roots}\}.$$

By Theorem 2, $\sum_{a,b} N(a, b) = q^2 M(q, d) = q^2$. The off-diagonal unordered pairs number $\binom{q}{2}$, by the previous paragraph. Hence

$$\sum_{a,b} N(a, b)^2 = q^2 + 2 \binom{q}{2} = q^2 + q(q-1) = q(2q-1),$$

and dividing by q^2 ,

$$\mathbb{E}[N^2] = \frac{q(2q-1)}{q^2} = \frac{2q-1}{q} = 2 - \frac{1}{q}. \quad \square$$

Corollary 7. *For every prime power q ,*

$$\text{Var}(N) = \mathbb{E}[N^2] - (\mathbb{E}[N])^2 = 1 - \frac{1}{q}.$$

In particular the conjectured variance $2 - 1/q$ is false for every $q > 1$.

Proof. By Theorem 2, $\mathbb{E}[N] = M(q, d) = 1$. Hence, by Lemma 6,

$$\text{Var}(N) = \left(2 - \frac{1}{q}\right) - 1^2 = 1 - \frac{1}{q}.$$

For $q > 1$ we have $1 - 1/q \neq 2 - 1/q$. Explicitly, $q = 3, 5, 11$ give variances $2/3, 4/5, 10/11$, whereas the conjecture claims $5/3, 9/5, 21/11$. $\square$

Remark 8. The conjectured variance $2 - 1/q$ is exactly the second moment $\mathbb{E}[N^2]$ of Lemma 6. Likewise, when $g = 1$ the conjectured “main term” $1 + (q-1)/q = 2 - 1/q$ is also exactly this second moment. So the conjecture has substituted the second moment both for the mean (up to the g -dependent factor) and for the variance.

5 The “main term” objection, and why it fails

A charitable reader might propose that $1 + (q-1)/q^{\gcd(d-1, q-1)}$ is asserted only “as the main term”, i.e. as an asymptotic leading term rather than an exact finite value, so that Theorem 2 does not refute it. This reading does not save the conjecture.

1. **The statement is about finite q , and the correction is not negligible.** The definition fixes a single finite field $\mathbb{F}_q$, and the formula is asserted for that q . Even under a leading-term reading, the leading term of a quantity that is identically 1 must be 1. The proposed correction $(q-1)/q^g$ is not a lower-order error: when $g = 1$ it equals $(q-1)/q > 1/2$ for all $q \geq 2$, which is $O(1)$ and cannot be absorbed into lower-order terms. An asymptotic claim with leading constant $1 + \Omega(1)$ is simply false for a quantity equal to 1.
2. **“Main term” cannot reconcile two contradictory clauses.** The conjecture asserts both the value $1 + (q-1)/q^g$ and, when $g = 1$, that the mean “is exactly 1”. At $q = 5$, $d = 2$ this reads $9/5$ versus 1. No interpretation of “main term” makes one number simultaneously equal to $9/5$ and to 1.
3. **The variance clause is separately false.** Independent of any reading of “main term”, the assertion “the variance is $2 - 1/q$ ” is plain arithmetic, and Corollary 7 shows the variance is $1 - 1/q \neq 2 - 1/q$ for every $q > 1$.

Therefore all three components fail: the exact mean is 1, not $1 + (q-1)/q^g$; for $g = 1$ the value $1 + (q-1)/q$ is the second moment, not the mean; and the variance is $1 - 1/q$, not $2 - 1/q$.

Summary

For every prime power q and every d : $M(q, d) = 1$; $\mathbb{E}[N^2] = 2 - 1/q$; $\text{Var}(N) = 1 - 1/q$. The conjecture’s main term, exact-mean clause, and variance clause are respectively wrong, internally contradictory, and wrong. Hence conjecture 00000001097 is **FALSE**.